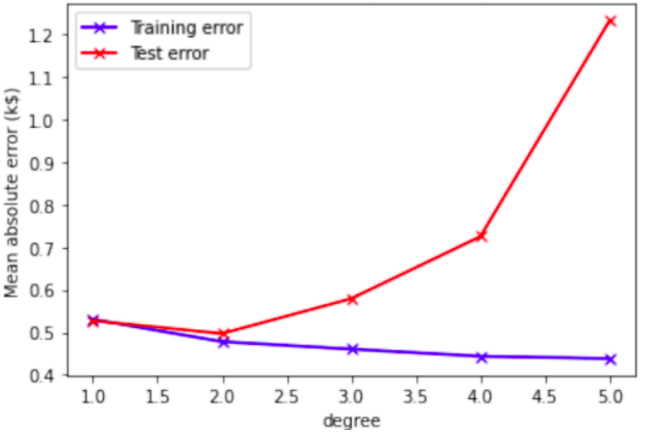
Week 6 | Solve With TAs

Q1) Following chart plots MAE against degree of polynomial features for regression task on California housing dataset. Which is the optimal value of degree



○ 1

○ 2

○ 3

○ 4

Solution:

From the chart:

- For degree 1, the test error is relatively low.
- At degree 2, the test error appears lowest before it starts to increase.
- For degrees 3 and above, the test error increases noticeably, suggesting overfitting.

Thus, the optimal degree is likely 2, as it provides the lowest test error while avoiding the overfitting seen at higher degrees.

Q2) For Ridge Regression, if the regularization parameter is very high, which options are true?

(A) Large coefficients are significantly penalized
(B) Can lead to a model that is too simple and ends up underfitting the data
(C) Large coefficients are not penalized
(D) Can lead to a model that is too simple and ends up overfitting the data

Solution:

For Ridge Regression, when the regularization parameter is very high:

- **(A) Large coefficients are significantly penalized:** True. A high regularization parameter in Ridge Regression heavily penalizes large coefficients, driving them closer to zero.
- **(B) Can lead to a model that is too simple and ends up underfitting the data:** True. When the regularization is too strong, the model may become overly simplistic, failing to capture important patterns in the data, leading to underfitting.
- **(C) Large coefficients are not penalized:** False. In Ridge Regression, large coefficients are explicitly penalized, especially with a high regularization parameter.
- **(D) Can lead to a model that is too simple and ends up overfitting the data:** False. Overfitting typically happens with low regularization, while high regularization encourages simpler models that may underfit rather than overfit.

So, the correct answers are **(A)** and **(B)**.

Q3) Ajay has a dataset that has a non-linear relationship between the features and the label. The dimension of training samples is 4×2 (excluding the additional all-ones feature for the bias term, which you should add), label vector is 4×1 and weight vector is 3×1. For all your calculations, consider SSE to be $\frac{1}{2}\|\mathbf{X}^T\mathbf{w} - \mathbf{y}\|^2$.

$\mathbf{X}$	$\mathbf{y}$
$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	1
$\begin{bmatrix} 2 \\ 2 \end{bmatrix}$	1
$\begin{bmatrix} 3 \\ 2 \end{bmatrix}$	2
$\begin{bmatrix} 4 \\ 3 \end{bmatrix}$	2

$\mathbf{w}$
0.1
0.2
0.3

i) Suppose Ajay has to compute the ridge loss value for this data with regularization parameter value as 0.2 for performing ridge regression. What will be the value of ridge loss correct up to 2 decimal places?

ii) For the same dataset, suppose Ajay has to compute the lasso loss value with regularization parameter value as 0.2 for performing lasso regression. What will be the value of lasso loss correct up to 2 decimal places?

Solution:

Matrix $\mathbf{X}_{d\times n}$ after adding bias term will look like

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 2 & 2 & 3 \end{bmatrix}$$

Predicted vector is

$$\widehat{\mathbf{y}} = \mathbf{X}^T\mathbf{w} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 3 & 2 \\ 1 & 4 & 3 \end{bmatrix} \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 1.1 \\ 1.3 \\ 1.8 \end{bmatrix}$$

SSE

$$\frac{1}{2}\|\mathbf{X}^T\mathbf{w} - \mathbf{y}\|^2 = \frac{1}{2}\|\widehat{\mathbf{y}} - \mathbf{y}\|^2 = \frac{1}{2}\left\| \begin{bmatrix} 0.6 \\ 1.1 \\ 1.3 \\ 1.8 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix} \right\|^2 \approx 0.35$$

The Ridge loss is given by

$$\text{ridge loss} = \frac{1}{2}\|\mathbf{X}^T\mathbf{w} - \mathbf{y}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|_2^2$$

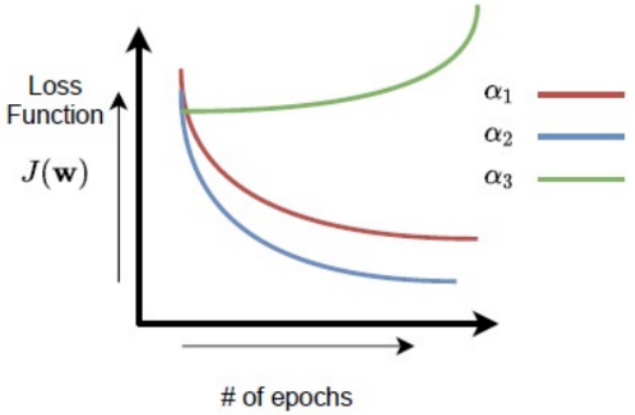
$$= 0.35 + 0.1(0.14) = 0.364$$

The Lasso loss is given by

$$\text{lasso loss} = \frac{1}{2}\|\mathbf{X}^T\mathbf{w} - \mathbf{y}\|^2 + \lambda\|\mathbf{w}\|_1$$

$$= 0.35 + 0.2(0.6) = 0.47$$

Q4) Brijmohan trains his linear regression model with three different values of learning rate, i.e. α_1, α_2 and α_3, for the same training set. He gets learning curves as displayed in the following chart.



Which of the following correctly explains relationship among α_1, α_2 and α_3?

a) $\alpha_1 > \alpha_2 > \alpha_3$
b) $\alpha_1 < \alpha_2 < \alpha_3$
c) $\alpha_2 > \alpha_1 > \alpha_3$
d) $\alpha_3 > \alpha_1 > \alpha_2$
e) $\alpha_1 > \alpha_3 > \alpha_3$

Solution:

1. Green Curve (α_3): This curve does not converge and instead diverges upwards, indicating that α_3 is too high. When the learning rate is too high, gradient descent overshoots the optimal point, causing the loss to increase rather than decrease.

2. Red Curve (α_1): This curve converges, but more slowly than the blue curve. This suggests that α_1 has a lower learning rate than α_2, leading to a slower decrease in the loss over epochs.

3. Blue Curve (α_2): This curve converges the fastest and reaches the lowest point in the shortest number of epochs, indicating that α_2 is the optimal learning rate among the three. It is not too high to cause divergence and not too low to slow down convergence.

From this analysis, the relationship among the learning rates is:

$$\alpha_1 < \alpha_2 < \alpha_3$$

Q5) For a parameter θ, there are two estimators θ_1 and θ_2 such that :

$$\left|\left|E[\theta_1] - \theta\right|\right|^2 > \left|\left|E[\theta_2] - \theta\right|\right|^2$$

Choose the correct option.

A. θ_1 is a better estimator for θ when compared to θ_2
B. θ_2 is a better estimator for θ when compared to θ_1
C. Cannot conclusively say anything with the given information.

Solution:

The given inequality doesn't account for the variance of the estimators.

Let us look at an example in $\mathbb{R}$. Let the true value of θ be 3.

Over 10 datasets, we get the following as the estimates for θ_1 and θ_2.

DATASET	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}
θ_1	3.5	3.7	2.9	3.3	3	3.9	2.5	3.1	3.1	3
θ_2	10	-10	5	6	-4	-7	8	-4	-4	3

$$\text{mean}(\theta_1) = \frac{3.5 + 3.7 + 2.9 + 3.3 + 3 + 3.9 + 2.5 + 3.1 + 3.1 + 3}{10} = \frac{31}{30} = 3.1$$

$$\text{mean}(\theta_2) = \frac{10 - 10 + 5 + 6 - 4 - 7 + 8 - 4 - 4 + 3}{10} = 3$$

Q7) To implement a five-fold cross-validation, we aim to identify the optimal model for a linear regression task. We are going to consider a standard linear regression, a polynomial kernel regression, and a Gaussian kernel regression. By the end of the cross-validation process, how many models will have been trained in total?

Solution:

Every $(\text{train set}, \text{ validation set})$ split is going to give a model.
At the end,we will have 5 models with usual linear regression, 5 models with polynomial kernel regression and 5 models with Gaussian kernel regression.

In total, we would have learned 15 different models.

Q8) Consider that the three weight vectors $\mathbf{w}_1, \mathbf{w}_2$ and $\mathbf{w}_3$ are learned for a six dimensional dataset using a linear regression model or regularized linear regression model (Not in any particular order).

$$\mathbf{w}_1 = \begin{bmatrix} 0.5 & 0 & 0.025 & 0 & 0 & -0.14 \end{bmatrix}^T$$

$$\mathbf{w}_2 = \begin{bmatrix} 0.8 & -0.23 & 0.45 & 0.2 & 0.31 & -0.54 \end{bmatrix}^T$$

$$\mathbf{w}_3 = \begin{bmatrix} 0.24 & -0.03 & 0.1 & 0.02 & 0.09 & -0.14 \end{bmatrix}^T$$

Solution:

To determine which of the weight vectors corresponds to a regularized model,we need to identify the weight vector with smaller coefficients (closer to zero) across all dimensions. Regularization methods (like Lasso or Ridge regression) typically reduce the magnitude of weights to avoid overfitting.

- $\mathbf{w}_2$ has the largest coefficients with significant values across multiple dimensions, suggesting it is likely from a **non-regularized model** like **LASSO** or **RIDGE**.
- $\mathbf{w}_1$ and $\mathbf{w}_3$ have smaller coefficients, suggesting they may be from a **regularized model**.
- Comparing $\mathbf{w}_1$ and $\mathbf{w}_3$, $\mathbf{w}_3$ has even smaller values across the dimensions, indicating it is the most regularized.

So, the order is likely:

- $\mathbf{w}_2$: Linear Regression
- $\mathbf{w}_1$: Regularized model i.e. Ridge since coefficients are close to zero but not exactly zero
- $\mathbf{w}_3$: More regularized model i.e. LASSO since more coefficients are exactly zero (e.g., with a stronger regularization parameter)

Q9) Consider the following dataset:

$$\mathbf{X} = \begin{bmatrix} -3 & 5 & 4 \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} -10 & 20 & 20 \end{bmatrix}^T$$

Assuming a ridge penalty $\lambda = 50$, what will be the value of $\dfrac{\widehat{\mathbf{w}}_{ridge}}{\widehat{\mathbf{w}}_{MLE}}$? Here $\widehat{\mathbf{w}}_{ridge}$ and $\widehat{\mathbf{w}}_{MLE}$ are the Ridge and MLE estimates of the weight vectors, respectively.

Solution:

Calculate the MLE (Maximum Likelihood Estimate):

The MLE for linear regression weights $\widehat{w}_{\text{MLE}}$ is given by:

$$\widehat{w}_{\text{MLE}} = (XX^T)^{-1}Xy$$

Here,$X = \begin{bmatrix} -3 & 5 & 4 \end{bmatrix}$ and $y = \begin{bmatrix} -10 & 20 & 20 \end{bmatrix}^T$

$$XX^T = (-3)^2 + 5^2 + 4^2 = 9 + 25 + 16 = 50$$

$$Xy = (-3)(-10) + (5)(20) + (4)(20) = 30 + 100 + 80 = 210$$

Therefore,

$$\widehat{w}_{\text{MLE}} = \frac{Xy}{XX^T} = \frac{210}{50} = 4.2$$

Calculate the Ridge Estimate:

The ridge estimate $\widehat{w}_{\text{ridge}}$ is given by:

$$\widehat{w}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$$

Since X is one-dimensional, we can simplify this as:

$$\widehat{w}_{\text{ridge}} = \frac{X^T y}{X^T X + \lambda} = \frac{210}{50 + 50} = \frac{210}{100} = 2.1$$

Calculate the Ratio:

Now, we find the ratio $\frac{\widehat{w}_{\text{ridge}}}{\widehat{w}_{\text{MLE}}}$:

$$\frac{\widehat{w}_{\text{ridge}}}{\widehat{w}_{\text{MLE}}} = \frac{2.1}{4.2} = 0.5$$

Q10) Consider a dataset with the following data points and the target variable:

Sample No	x	y
1	3	8
2	0	3
3	5	12
4	6	13

The linear regression model is given by $y = w_0 + w_1 x$. Assume that the Leave-One-Out Cross-Validation technique is applied. Based on the above information, answer the following questions

i) Enter the value of w_1 obtained when the 3rd sample is used as the validation data point.

ii) What will be the predicted value for the left-out data point?

Solution:

To apply Leave-One-Out Cross-Validation with the 3rd sample as the validation point, we fit the linear regression model based on the remaining three data points.

Set up the Training Data (Excluding the 3rd Sample)

When we leave out the 3rd sample (where $x = 5$ and $y = 12$), the remaining samples are:

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 0 & 6 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 8 \\ 3 \\ 13 \end{bmatrix}$$

where each column of $\mathbf{X}$ represents $\begin{bmatrix} 1 & x_i \end{bmatrix}^T$ for a data point, and y represents the corresponding y-values.

Computing $\mathbf{w}$ using normal equations:

$$\mathbf{w} = \left(\mathbf{X}\mathbf{X}^T\right)^{-1}\mathbf{X}\mathbf{y}$$

$$\mathbf{w} = \left(\begin{bmatrix} 1 & 1 & 1 \\ 3 & 0 & 6 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix}\right)^{-1}\begin{bmatrix} 1 & 1 & 1 \\ 3 & 0 & 6 \end{bmatrix}\begin{bmatrix} 8 \\ 3 \\ 13 \end{bmatrix}$$

$$\mathbf{w} = \begin{bmatrix} 3 \\ \frac{5}{3} \end{bmatrix}$$

Thus $\mathbf{w}_0 = 3$ and $\mathbf{w}_1 = \frac{5}{3}$.

Predict the Value for the Left-Out Data Point (3rd Sample)

The left-out data point has $x = 5$. Using the model $y = 3 + \frac{5}{3}x$:

$$\widehat{y} = 3 + \frac{5}{3} \cdot 5$$

$$\widehat{y} \approx 11.33$$